

POAMS: LECTURE 5

UNIFORM MOTION AND TIME-DILATION

Having presented the Normal Realist alternative for Einstein's second postulate of Special Relativity, which remains as a postulate in General Relativity, I will now present the Normal Realist analogue of Special Relativity, which concerns only uniform motion. I will show how the time-dilation formula of Special Relativity can be derived in a simple way, without having to interpret the constant c as 'the speed of light in vacuo'. This will lead to Viv's 'cone model' of time-dilation. The lecture is based on some of 'Light Speed...' Chapters 2 and 3.

1. Uniform (linear) motion

Special Relativity, which deals exclusively with *uniform motion*, was first proposed by Einstein in 1905. Primarily, this was in response to a crisis in classical physics that had developed during the latter half of the 19th century, involving, among other things, Maxwell's equations, the wave theory of light, the proposal for the luminiferous ether, and the null results of the Michelson and Morley experiment. [*These were discussed in the last lecture*].

Essentially, Einstein's basic postulates are that the laws of physics are the same in all frames of reference which move *uniformly* (i.e. have constant velocity) relative to each other and that the 'speed of light in vacuo', c , is the same relative to all such frames. Frames of reference in relative uniform motion are known as *inertial frames*. Einstein's first postulate is known as the *Principle of Relativity* and implies that there is no physical experiment that can detect the difference between two inertial frames. Note that by constant velocity we mean constant speed in a fixed direction, so that inertial frames move along a *linear* path. Constant speed alone does not guarantee constant velocity. For example, a particle travelling at constant speed along a circular path does not have constant velocity.

Our common experience tells us that it does not make sense to talk of *absolute* motion, as though an object can be thought of as moving or stationary in itself, relative to nothing else, in the way that Newton had envisaged. As Mach insisted, only motion *relative* to a particular frame of reference has any logical meaning. For example, when we drive a car, the

speedometer tells us the speed of the car relative to the road. The road is stationary relative to Earth, but Earth is orbiting the Sun, which, in turn, is orbiting the galactic centre, and so on.

The fact that *mechanical* laws must be the same in all inertial frames was initially recognised by Galileo Galilei (1564-1642). In his famous treatise, 'Dialogue Concerning the Two Chief World Systems – Ptolemaic and Copernican' Galileo argued that a rock dropped from the top of the mast of a ship always lands at the foot of the mast regardless of whether the ship is stationary in a harbour or at sea, sailing at a constant velocity. In this way, he argued, the fact that freely falling objects always travel vertically does not imply that Earth is stationary. In the same treatise, he implies that the results of any mechanical experiment performed in the cabin of a ship would be the same, whether that experiment was performed with the ship stationary in a harbour or with the ship sailing at a constant velocity.

This fact that the laws of mechanics must be the same in all inertial frames was also appreciated by Newton. He stated:

'The motions of bodies included in a given space are the same among themselves, whether that space is at rest, or moves uniformly forwards in a right (straight) line without any circular motion.'

But why is the condition of *uniform* motion the important ingredient in all this? In contrast to relative uniform motion, non-uniform motion is very easy to detect. For example, let us imagine ourselves sitting in a jet plane, which travels at constant velocity relative to the ground. Suppose that the walls of our cabin are sound-proof and that shutters cover its windows. Any mechanical experiment we set up in the plane will not tell us whether the plane is in flight or standing stationary on the runway. It is only when we are able to look out of the window and are close enough to the ground that we can see we are travelling relative to the earth. However, when the plane encounters turbulence or decelerates on landing, we can tell that we are no longer in a uniformly moving frame, due to the sensations that we feel with the jolting and/or severe braking on touchdown. In this case, a mechanical experiment would be able to detect the non-uniform motion. In the same way, if we are riding on a child's spinning roundabout at constant speed, we can tell we are in non-uniform motion caused by the circular motion, due to the visceral effects we feel.

Since all inertial frames are equivalent as far as the laws of mechanics are concerned, it follows logically that they should also be equivalent as far as *all* the laws of Physics are concerned, since there is no reason to suppose that mechanics has a privileged position in Physics. It follows that our Normal Realist first postulate for uniform motion is the same as Einstein's first postulate for Special Relativity, namely:

Postulate 1: *The laws of Physics are the same in all inertial frames.*

It is important to realise here that this postulate does not imply that the laws of Physics are the same relative *only* to inertial frames. Indeed, it is incumbent upon any physical theory to ensure that, ultimately, its laws are the same for *all* observational frames. Einstein accomplished this when he formulated his General Relativity, which builds upon Special Relativity and incorporates non-uniform motion. Like Einstein, we begin with uniform motion, just as Galileo and Newton did before Einstein. This is for the reasons outlined above and because, from a mathematical point of view, it is the simplest type of motion to study.

The fact that all inertial frames are equivalent implies that even if a frame of reference at 'absolute rest' existed, it would be essentially undetectable. In realising this Einstein solved the perceived problem of the null results of the Michelson and Morley experiments: even if the 'ether' existed, it could never be detected. Our Normal Realist philosophy goes further: there can be no unempirical 'ether' frame of absolute rest in a truly relativistic theory. Newton, on the other hand, maintained his belief in a frame of reference at absolute rest, *i.e.*, the space through which the stars and planets move. Since such a frame of reference cannot be distinguished from any other inertial frame by experiment, Newton's view, from the outset, attracted scepticism from philosophers such as Gottfried Leibniz (1646-1716) and Bishop Berkeley.

Although this view of Newton's is clearly unsatisfactory, the trouble with insisting on the complete equivalence of all inertial frames is that it leaves us without a universal reference frame relative to which we can define all other inertial frames. Mach argued that since it is the various interactions between all matter in our 'universe' (over and above any 'gravitational' effects) which are the primary source of inertial effects, and that the bulk of the matter resides in what he termed the 'fixed stars', then any inertial frame is a frame of reference in some privileged state of uniform motion

relative to the average motion of that stellar system. For Mach, it is this holistic system, through its masses, distributions and motions, which determines, both locally and non-locally, an inertial frame. This view is essentially known as *Mach's Principle*.

It could be argued that a more satisfactory approach is based on Newton's First Law, since this enables us to define an inertial frame without reference to other such frames. Newton's First Law states that,

'A body, not subjected to any force, remains at rest or continues to move in a straight line with constant speed'.

Since a frame of reference in uniform motion is not subjected to any Newtonian force, we are led to the following definition:

*An **inertial frame** is any frame of reference in which spatial relations, as determined by rigid scales at rest in the frame, are Euclidean and where there exists a measure of duration in terms of which freely moving particles remain at rest or continue to move in straight lines at constant speed.*

Using this definition, we can detect, at least in theory, whether or not we are in an inertial frame.

2. The Normal Realist approach to uniform motion

To recap, it follows from this lecture and the last, that our Normal Realist postulates for the study of *uniform motion* are:

Postulate 1: *The laws of Physics are the same in all inertial frames.*

Postulate 2: *Observational distance and time have a constant ratio of units, c , for all inertial observers sharing that same conventional choice of units.*

These postulates express a purely observational, or phenomenalist, approach to Physics, of the kind adapted by Mach. Our first postulate is the same as that of Einstein for Special Relativity. Our second postulate is our phenomenalist alternative to Einstein's conceptually redundant second postulate that the 'speed of light *in vacuo*', c , is the same for all inertial

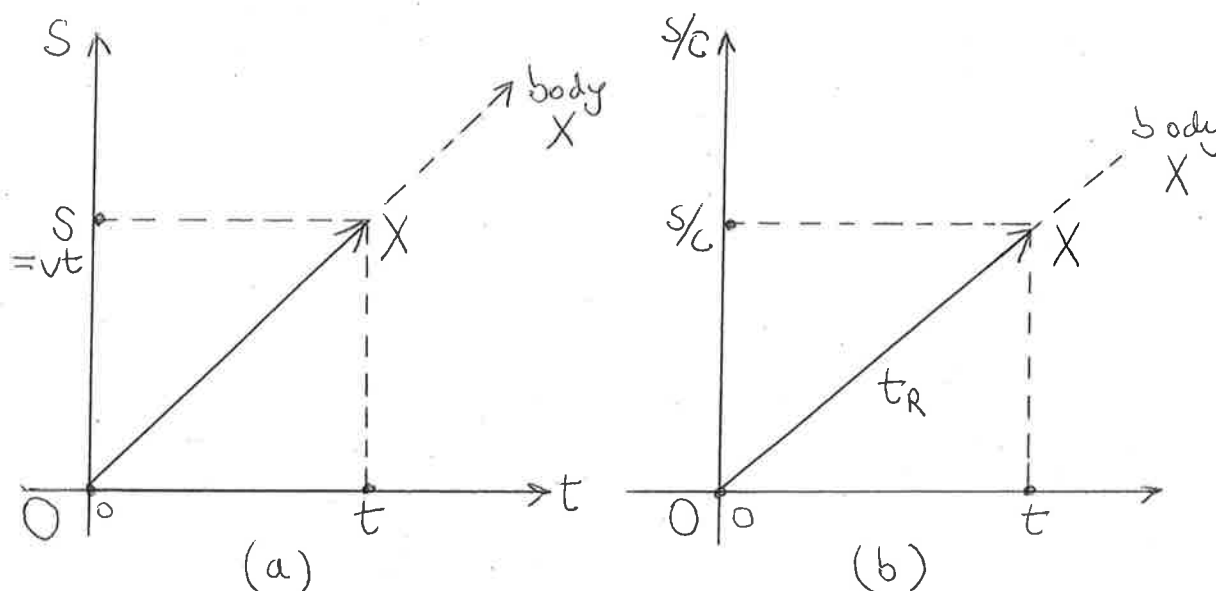
observers. Since the laws of electromagnetism must be the same relative to all inertial observers by Postulate 1, it follows that the constant, c , appearing in Maxwell's equations must be the same for all observers, leading to our second postulate.

In contrast to Einstein's, all that our second postulate states is that c acts as a *conversion factor* for interrelating observer-projected measures of distance and time. Hence, for example, if the unit of distance is chosen to be the metre (m) and the unit of time is chosen to be the second (s), so that $c \cong 3 \times 10^8 \text{ m s}^{-1}$, then our second postulate states that a distance of one metre may be recorded as $1/3 \times 10^{-8}$ seconds relative to all inertial observers. Here, the 'second' is being used in two different senses; one in the usual sense of *duration* and the other in the sense of a distance equivalent. This idea should not appear the least bit strange. These days, measuring a journey by the time it takes is often more informative than the number of miles or kilometres it may be. But of course, different travellers can take different times to complete the same distance. What our Postulate 2 states is that the *observational* distance of one metre is always equivalent to an *observational* time of $1/3 \times 10^{-8}$ seconds.

In our approach to relative uniform motion, as in Special Relativity, the role of the observer is key. Note that in large-scale relational systems of inertial frames, there is no uniquely *self-centred* observational frame of reference. Nor are those other 'viewpoints' necessarily those of other *human* observers. They may be the reference frames of instruments of observation, and any object whatsoever can be an instrument in this sense and therefore a rudimentary 'observer'. It is in terms of this informational *logos* that objects, at various levels of simplicity or complexity, and various levels of automaticity or autonomy, communicate their relative existences and spatial interrelations to one another. In this way, each observer becomes a 'reader' of the informational script, compiled by other observers, both complex and rudimentary, and the world is the story they tell one another in all this prolific correspondence. All this is consistent with the stance of phenomenalism insofar as it eschews all discussion of any supposed 'reality' lying behind and beyond *observation*, in the sense we have defined.

3. Time-dilation in relative uniform motion

In standard representations of the motions of bodies, the distance of a body, X , relative to some fixed origin, O , is described by a curve plotted against two perpendicular axes, labelled s and t , which signify distance and time respectively. If the motion of X relative to O is uniform and X passes O at time $t = 0$, then this curve is a straight line which passes through the origin with constant gradient $v = s/t$. The equation of this straight line is then $s = vt$, as shown in Figure (a) below.



Of course, v , here, is the constant speed of X relative to O and s is the distance travelled by X from O in the time period t , relative to either O or X , since in classical Physics, the passage of time is supposed to be the same for all observers.

However, let us now think of t as the passage of time recorded by a 'clock' moving with the body X . This is the natural choice for the duration of the motion since it is *independent of the observer*. Here, a 'clock' represents any device capable of recording a passage of time, which includes biological as well as mechanical devices. Note that in Figure (a), there is no special significance attached to the length of the line with equation $s = vt$ for a given value of t , since in this kind of representation there is no fundamental interconnection between distance and time as such.

Taking account of our Postulate 2, this situation is drastically altered. If c is treated as a conversion factor, then a fixed geometrical relationship is established between the dimensions of s and t . Hence, on the grounds of

plain common sense, it is to be expected that with this overall connection, motion will affect time in some way. This is because motions are now geometro-dynamical combinations of time measures in different dimensions – analogous to the lengths of an ordinary area. Motion, that is to say, is now a composite space-time extension measured in units of approximately 3×10^8 metres/ c or seconds, which are numerically and dimensionally the same. Remember here that the motion of the body X is the motion of X relative to the specified observer at O , there being no single, universal ‘God’s-eye-view’ of how motion separating O and X can be seen simultaneously through the eyes of an observer at O and an observer at X .

To reiterate, X travels a distance s measured in metres, in a time t , recorded in seconds by X ’s clock, relative to O . In keeping with our Postulate 2, this distance travelled by X relative to O can be recorded in seconds by the use of the conversion factor c . Then the distance-time diagram, Figure (a) above, becomes the *two-dimensional time* diagram, Figure (b) above. In Figure (b), the two axes (labelled s/c and t) are both measured in units of seconds. Bearing in mind that the line segment OX represents the path of X ’s clock relative to O , it follows that the length of OX in this second graph is the time, t_R , as recorded by O , for X ’s clock to reach its position relative to O after time t . Then t_R will be simply the length of the hypotenuse of the triangle in Figure (b), given by Pythagoras’s Theorem as

$$t_R^2 = t^2 + (s/c)^2 \quad (1)$$

Now the *speed*, v , of X relative to O (that is, relative to O ’s clock) is given by $v = s/t_R$, so that equation (1) becomes

$$\begin{aligned} t_R^2 &= t^2 + (vt_R/c)^2 \\ \therefore t^2 &= (1 - v^2/c^2)t_R^2 \\ \therefore t &= (1 - v^2/c^2)^{1/2} t_R \end{aligned} \quad (2)$$

The positive square root is taken in equation (2) due to considerations of causality. This equation implies that O ’s time is dilated relative to X ’s time and that a moving clock runs *slower* relative to a stationary observer’s clock. Note also that equation (2) indicates that v cannot be greater than c ,

since otherwise, t^2 would have to be negative! In this case, there is no sensible question of there being any 'superluminal' speeds. Equation (2) also indicates, once again, that c cannot be treated as if it were a speed, since otherwise for $v = c$, $t = 0$, independently of t_R . In this case, the 'travelling' clock would register no time difference, regardless of how great the distance 'travelled' might be.

Equation (2) is the standard formula for *time-dilation* which appears in Special Relativity. However, our derivation of equation (2) is much shorter than the derivation in Special Relativity, where it arises as a consequence of the Special Lorentz Transformation. In contrast, our derivation puts time-dilation right 'at the heart' of relative uniform motion.

Our derivation shows that, observationally, there are two distinct measures of the speed of relative motion. One is the rate of change of observational distance relative to the time, t_R , as registered by the observer of the motion, known as the **observational time**, and the other is the rate of change of that same distance relative to the time, t , that the observer sees in his telescope/spectroscope registered by the moving body itself, known as the **proper time**. It also shows that time is a relative rather than an absolute concept as envisaged by Newton. It is important to re-emphasise here, however, that any moving clock records the proper time, t , relative to its own rest frame, so that the proper time is *independent* of any observer. In contrast, the observational time, t_R , is clearly observer dependent.

4. Experimental Evidence for time-dilation

One of the most famous verifications of the time-dilation effect concerns 'elementary particles' known as *muons*. These are produced in appreciable numbers in the upper atmosphere by the primary cosmic radiation entering from outer space and have an intrinsic mean lifetime of 2.2×10^{-6} seconds. Muons have a speed of approximately $0.9965c$ relative to an observer on Earth, so that according to classical physics, the mean distance they could cover before decaying is about 650 metres. In other words, classical physics predicts that the high-altitude muons should decay long before they reach sea level. However, muons are, in fact, found in significant numbers at sea level, as first detected in experiments performed by Bruno Rossi (1905-1993) and David Hall in 1941. The explanation is that since the muons, with their speed of $0.9965c$, have a proper mean lifetime of

$t = 2.2 \times 10^{-6}$ seconds, Equation (2) gives them a mean lifetime of $t_R \approx 2.64 \times 10^{-5}$ seconds, relative to a terrestrial observer. Hence, the high-altitude muons are able to travel a mean distance of about 7.9 kilometres relative to Earth before decaying, and so can easily reach sea level, as observed.

More recently, experiments performed by Joseph Hafele (1933-2014) and Richard Keating (1941-2006) in 1971, using atomic clocks on board commercial airliners, also verified the time-dilation formula. But perhaps by far the most convincing demonstration lies in the Stanford Linear Accelerator, which accelerates 'elementary particles' up to speeds which are significant fractions of c . This accelerator is over three kilometres long and cost a hundred million dollars to build, whereas if classical physics were correct, it would have to only about three centimetres long! (This, of course, is because, as in the case of muons, due to time-dilation, the distance travelled by the high-speed elementary particles before they decay is much further than that predicted by classical physics).

5. The cone model

The idea of using a right-angled triangle to deduce the time-dilation effect has been used by others, notably by Bondi. However, our derivation of Equation (2) provides a *model* of this phenomenon. Let t , s/c and t_R all be treated as variables in Equation (1). Then Equation (1) is the equation of a right-circular *cone* in three dimensions, relative to three mutually perpendicular time axes, t , s/c and t_R , which record proper time, observational distance-time and observational time respectively. As such, this conic surface represents a three-dimensional graph of time-dilation, as shown on the front cover of 'Light Speed ...'. I have kept the original model of the cone, built by Viv using plaster. The axis of the cone is the t_R axis and its apex is situated at (0, 0, 0). This model illustrates effectively the relationship between any given values of t , s/c and t_R and clearly indicates that $t < t_R$ for $v \neq 0$. The larger the value of s/c , the greater is the difference between t and t_R .

For any fixed value of t_R , Equation (1) is the equation of a *circle* of radius t_R , centred at the origin. Such circles are obtained from the *end elevation* of the cone. On the other hand, if t is fixed and the time-measure s/c is

plotted against the time-measure t_R , the result is a *rectangular-hyperbola* with equation

$$t_R^2 - (s/c)^2 = t^2$$

Such hyperbolæ are obtained from the *side elevation* of the cone. Any such curve is asymptotic to the lines with equations $s = \pm ct_R$. In mainstream Special Relativity, these lines form what is known as the (degenerate) *light cone* and represent ‘paths of light signals’ passing O at $t_R = 0$. Of course, there is no such physical interpretation of these lines in our Normal Realist approach.

In the next lecture, I shall present some of the consequences of the time-dilation formula and the associated ‘cone model’.

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